

■ A.2 Implementing Mathematical Algorithms

■ A.2.1 Runge-Kutta Methods

Runge-Kutta methods are used to find numerical approximations to the solutions of ordinary differential equations. The version we give here can be used to solve first-order equations of the form $y' = f(x, y)$. An important feature of the *Mathematica* implementation of Runge-Kutta methods is that you can give the function $f(x, y)$ in an explicit symbolic form.

```
RungeKutta::usage = "RungeKutta[f, {x, y}, {x0, y0}, h, ntot] gives
    ntot {x, y} pairs obtained by integrating the differential
    equation y'==f by the fourth order Runge-Kutta method."

RungeKutta[f_, {x_, y_}, {x0_, y0_}, h_, ntot_] :=
    NestList[RKStep[f, {x, y}, #, h]&, {x0, y0}, ntot]

RKStep[f_, {x_, y_}, {xn_, yn_}, h_] :=
    Block[{k1, k2, k3, k4},
      k1 = h f /. {x->xn, y->yn} ;
      k2 = h f /. {x->xn+h/3, y->yn+k1/3} ;
      k3 = h f /. {x->xn+2h/3, y->yn-k1/3+k2} ;
      k4 = h f /. {x->xn+h, y->yn+k1-k2+k3} ;
      { xn + h, yn + (k1 + 3k2 + 3k3 + k4)/8 }
    ]
```

Runge-Kutta methods.

The Runge-Kutta method works by taking a sequence of steps. The function `RKStep` does one step. `RungeKutta` itself applies `RKStep` many times using the functional iteration operation `NestList`.

Read in the package that implements
Runge-Kutta methods.

```
In[1]:= <<RungeKutta.m
```

This finds a numerical approximation to the solution of the equation $y' = 2y$. As expected, the result approximates an exponential in x .

```
In[2]:= RungeKutta[2 y, {x, y}, {0, 1}, .1, 5]
Out[2]= {{0, 1}, {0.1, 1.2214}, {0.2, 1.49182},
         {0.3, 1.82211}, {0.4, 2.22552}, {0.5, 2.71825}}
```

This generates a numerical approximation to the solution of the equation $y' = 1 - xy^2$, with $y(0) = 0$.

```
In[3]:= s = RungeKutta[1 - x y^2, {x, y}, {0, 0}, .1, 40];
```

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

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Here is a plot of the solution.

